

Time Series Analysis

Stochastic Processes – Time Series

A **stochastic process** is a collection of time indexed random variables defined on some space

(1) For a fixed $t \longrightarrow$

is a random variable.

(2) For fixed $\longrightarrow$

is a realization or sample function

In the case when

or

this **STOCHASTIC PROCESS** is called a **TIME SERIES**

Time series in sciences

Examples:

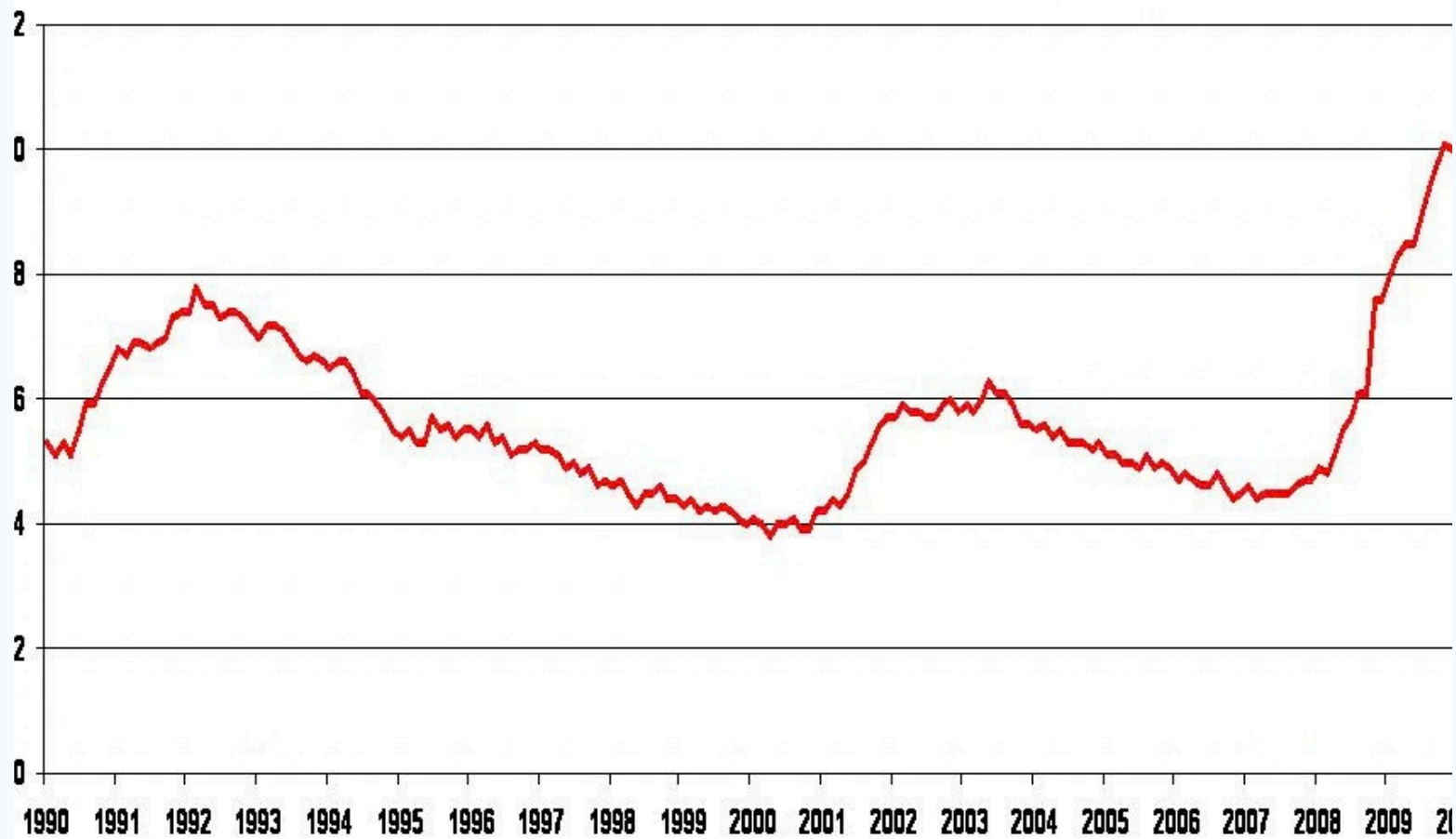
- In Economics, daily closing stock prices, weekly interest rates, monthly price indices and yearly earnings.
- In Meteorology, hourly wind speed, daily temperature and annual rainfall
- In Social Sciences annual birth rates, mortality rates, accident rates and various crime rates.

Ctd...

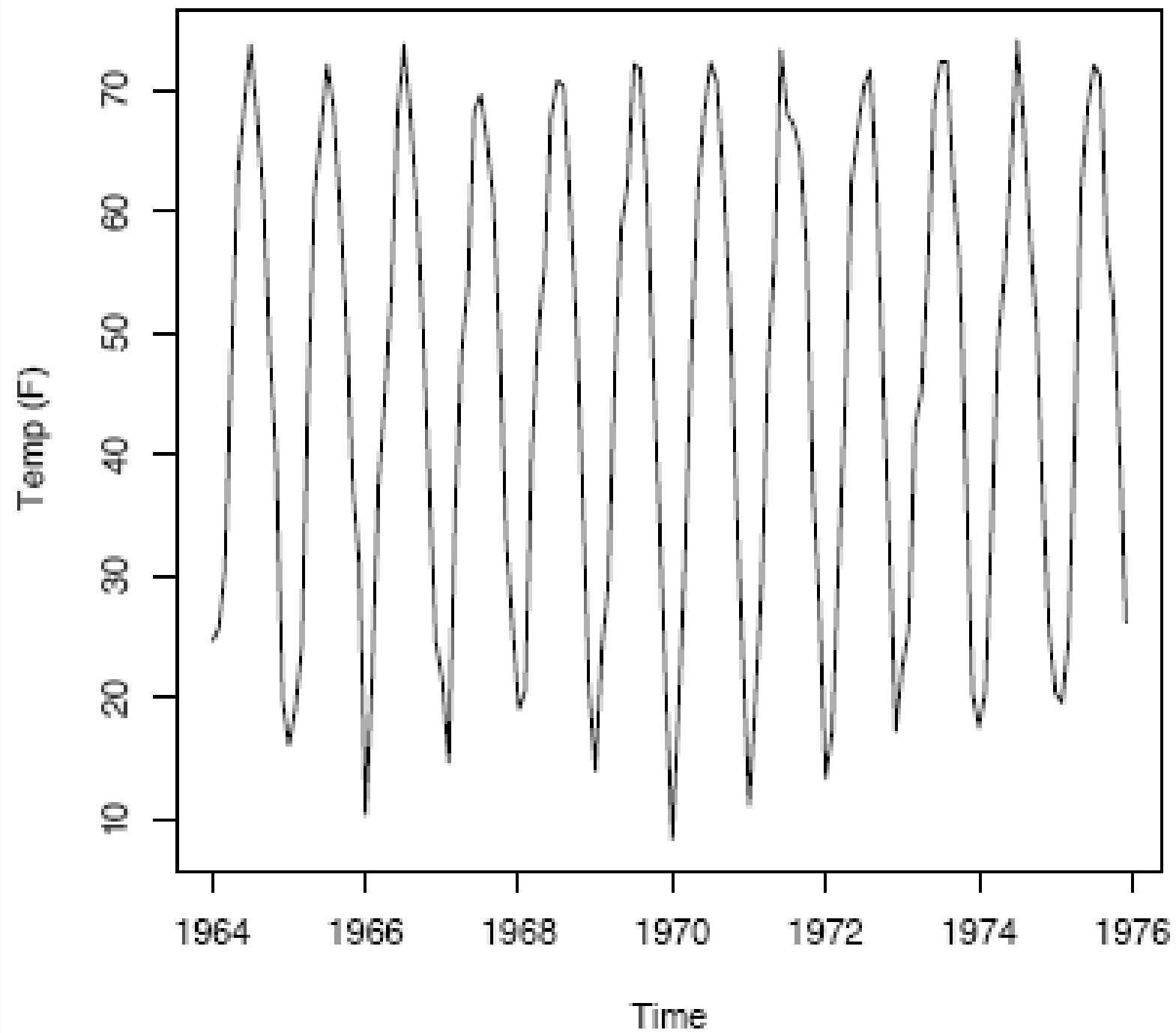
EXAMPLES

- Data in business, economics, engineering, environment, medicine, earth sciences, and other areas of scientific investigations are often collected in the form of time series.
- Hourly temperature readings
- Daily stock prices
- Weekly traffic volume
- Annual growth rate
- Seasonal ice cream consumption
- Electrical signals

Figure 2: The US Unemployment Rate since 1990



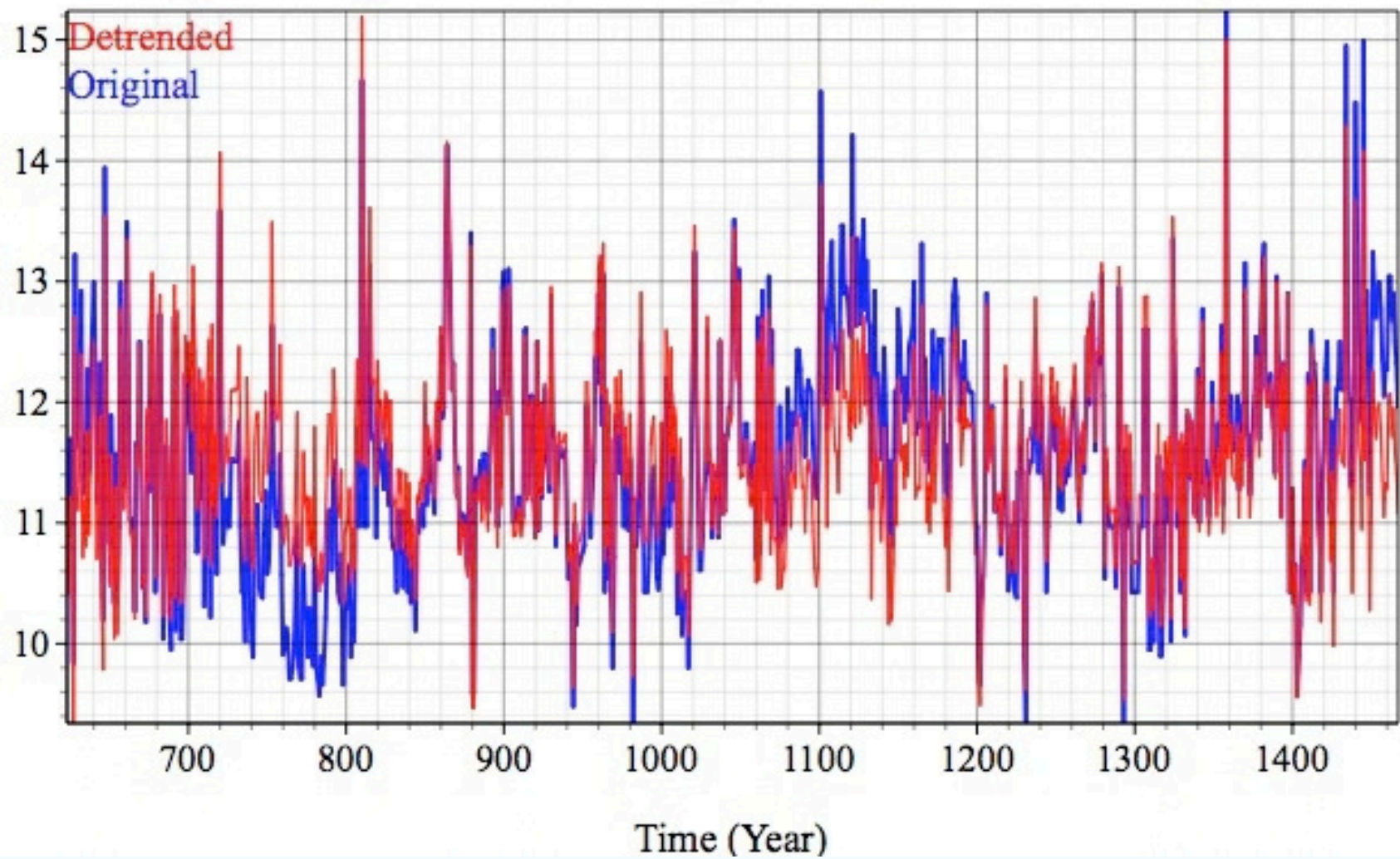
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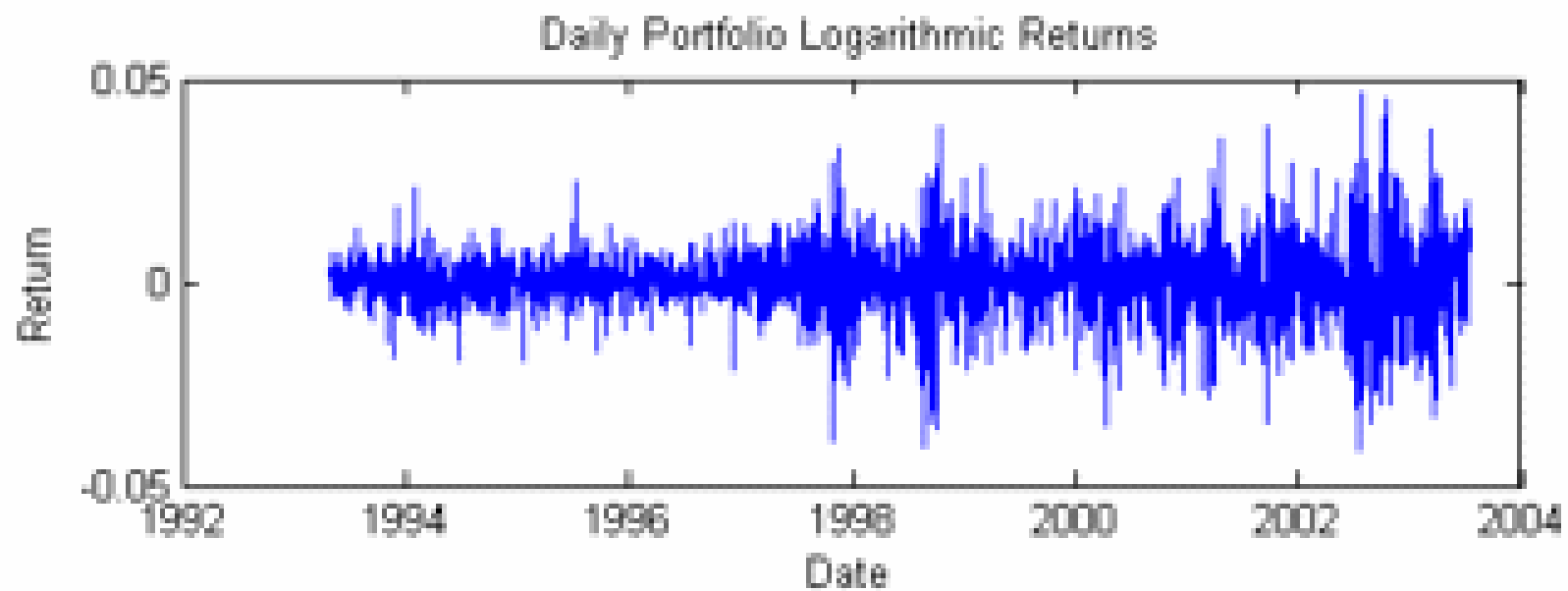


US Producer Price Index 12 Months Change - Finished Goods



Detrended Nile River Record





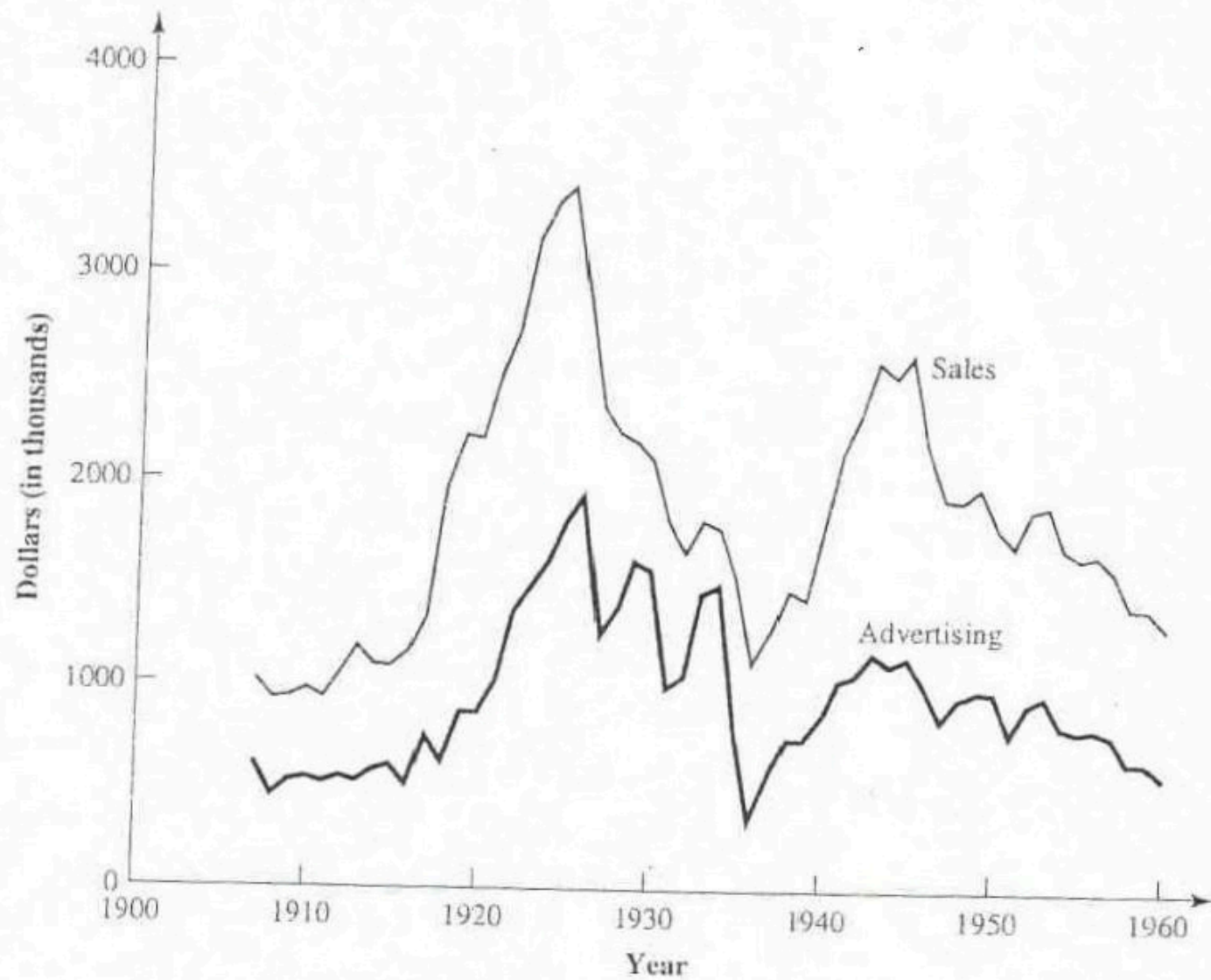
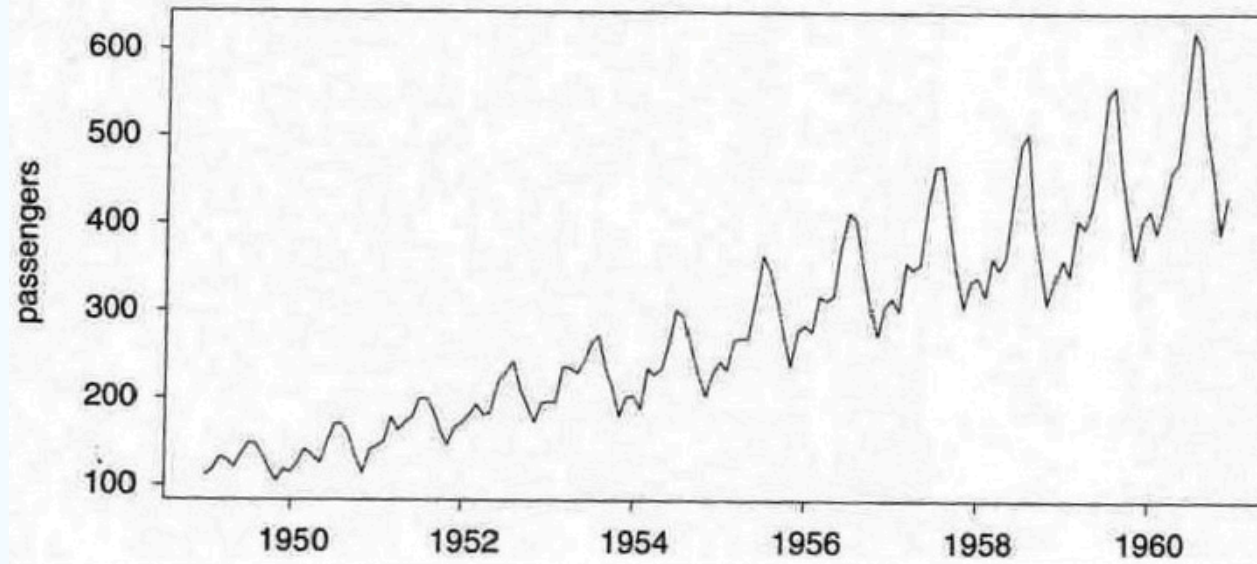
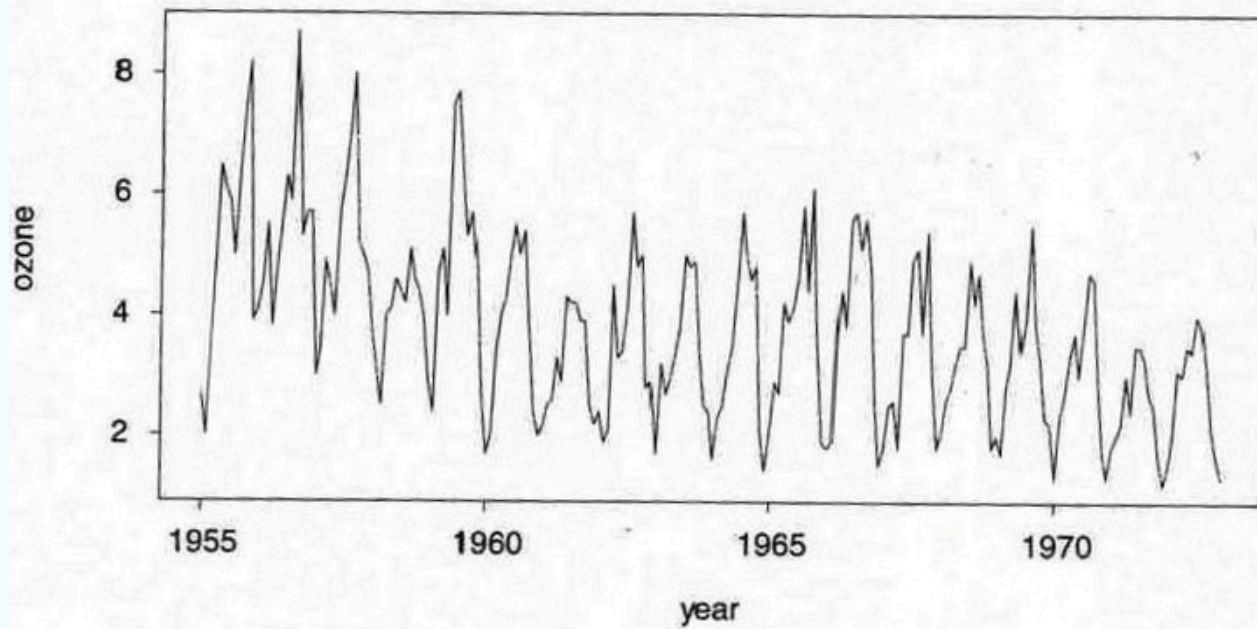


FIGURE 1.2 The Lydia Pinkham advertising and sales data.

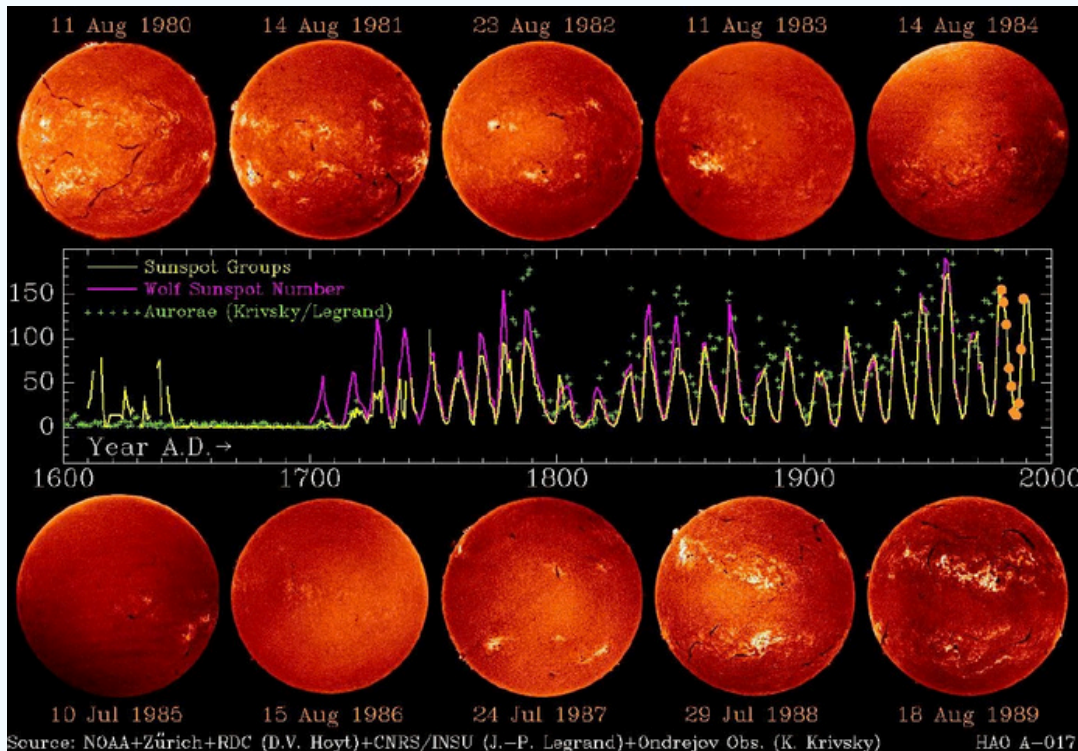
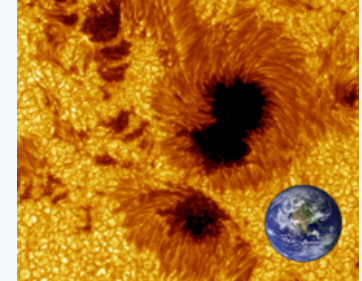
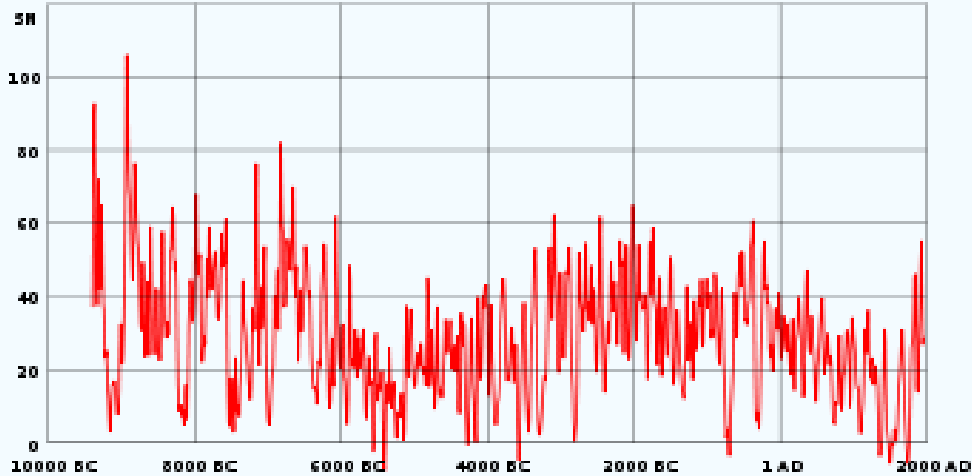
(a) International Airline Passenger Totals - Box-Jenkins Series G



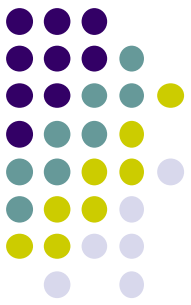
(b) Monthly Readings of Ozone at Downtown L.A. 1955-1972



SUNSPOT NUMBERS



The Wolf number (also known as the International sunspot number, relative sunspot number, or Zürich number) is a quantity that measures the number of sunspots and groups of sunspots present on the surface of the sun. The Sunspot Number is a crucial tool used to study the solar dynamo, space weather and climate change.



- **In Engineering electric signals, voltage and sound.**
- **In Geophysics ocean waves, earth noise in an area.**
- **EEG and EKG tracings in Medicine.**
- **In Agriculture annual crop production prices.**

Expectation, Variance, Autocovariance, Autocorrelation Functions

f_t = density function of Z_t

Autocovariance and Autocorrelation Functions of stationary process

Definition: (Weakly) **stationary process**:

Properties:

Properties of the autocorrelation function

Partial Autocorrelation Function (conditional correlation)

Correlation between two random variables that are k periods apart
removed the in-between linear dependence (between t and $t+k$).

Partial Autocorrelation Function (conditional correlation)

Motivation Think about a regression model
(without loss of generality, assume that $E(Z)=0$)

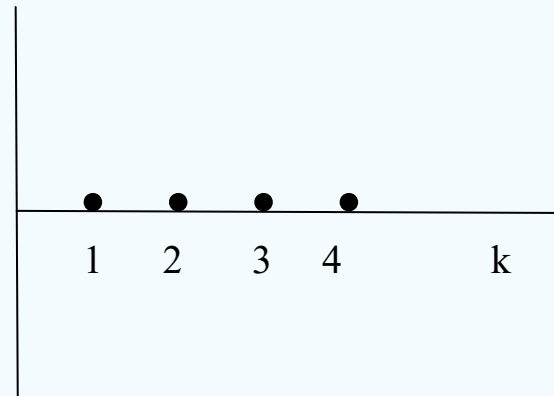
Dividing by the variance of the process:

Yule-Walker
equations

Examples of stochastic processes

E1: White Noise Process

A sequence of uncorrelated random variables is called a white noise process.



Objectives of Time series Analysis

- To understand the variability of the time series.
- To identify the regular and irregular oscillations of the time series.
- To describe the characteristics of the oscillations.
- To understand physical processes that give rise to each of these oscillations.
- **Forecasting of future observations**

STEPS IN TIME SERIES ANALYSIS

- Model Identification
 - Time Series plot of the series
 - Check for the existence of a trend or seasonality
 - Check for the sharp changes in behavior
 - Check for possible outliers
- Remove the trend and the seasonal component to get stationary residuals.
- Estimation
 - MME
 - MLE
- Diagnostic Checking
 - Normality of error terms
 - Independency of error terms
 - Constant error variance (Homoscedasticity)
- Forecasting
 - Exponential smoothing methods
 - Minimum MSE forecasting

CHARACTERISTICS OF A SERIES

- For a time series

THE MEAN FUNCTION:



THE VARIANCE FUNCTION:

The expected value of the process at time t .

CHARACTERISTICS OF A SERIES

- THE AUTOCOVARIANCE FUNCTION:



Covariance between the value at time t and the value at time s of a stochastic process Y_t .



The correlation of the series with itself

EXAMPLE

- **Moving average process:** Let ϵ_t i.i.d.(0, 1), and

$$X_t = \epsilon_t + 0.5 \epsilon_{t-1}$$

EXAMPLE

- **RANDOM WALK:** Let $e_1, e_2, \dots$ be a sequence of i.i.d. rvs with 0 mean and variance σ^2 . The observed time series

is obtained as

A RULE ON THE COVARIANCE

- If $c_1, c_2, \dots, c_m$ and $d_1, d_2, \dots, d_n$ are constants and $t_1, t_2, \dots, t_m$ and $s_1, s_2, \dots, s_n$ are time points, then

WEAK STATIONARITY

From, now on, when we say “stationary”, we imply weak stationarity.

EXAMPLE

- Consider a time series $\{Y_t\}$, where

$$Y_t = e_t$$

and $e_t \sim i.i.d.(0, \sigma^2)$. Is the process stationary?

Answer

EXAMPLE

- MOVING AVERAGE: Suppose that $\{Y_t\}$ is constructed as

and $\epsilon_t \sim i.i.d.(0, \sigma^2)$. Is the process $\{Y_t\}$ stationary?

Answer

EXAMPLE

- RANDOM WALK

where $\epsilon_t \sim i.i.d.(0, \sigma^2)$. Is the process $\{Y_t\}$ stationary?

Answer

EXAMPLE

- Suppose that time series has the form

where a and b are constants and $\{e_t\}$ is a weakly stationary process with mean 0 and autocovariance function k . Is $\{Y_t\}$ stationary?

Answer

EXAMPLE

where $\epsilon_t \sim i.i.d.(0, \sigma^2)$. Is the process $\{Y_t\}$ stationary?

Answer

1.3 Components of a time series

A time series is a collection of data obtained by observing a response variable at periodic points in time. X_t is used to denote the value of the variable x at time t .

A time series is made up of one or more components of the following :

- **1. Trend**
 - **Measures the average change in the variable per unit time. In other words it measures the change in the mean level of the time series.**

Ctd...

2. Seasonal variations

- Periodic variations that occur with some degree of regularity within a year or shorter.

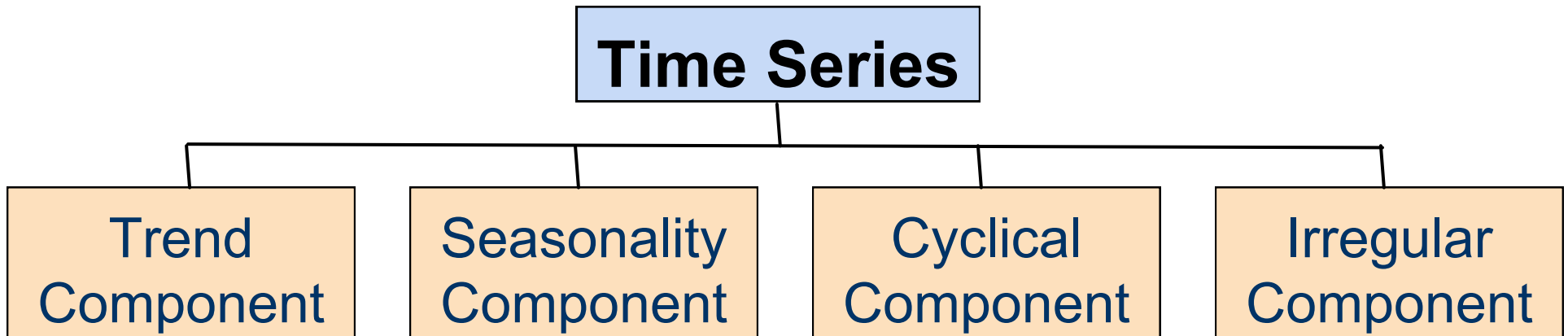
- **3. Cyclical variations**

- Recurring up and down movement which are extended over a long period.(usually 2 years or more)

- **4. Irregular variations**

- Random Fluctuations which happen due to errors.

Time-Series Components



Time series models can be classified into two types:
Additive model & multiplicative model

The four components are combined in the additive time series model & multiplicative model as follows:

$$Y_t = T + S + C + R \text{ (additive)}$$

$$Y_t = T * S * C * R \text{ (multiplicative)}$$

Where Y_t – the actual value

T – trend or long-term movement

S – seasonal movement

C – cyclical movement

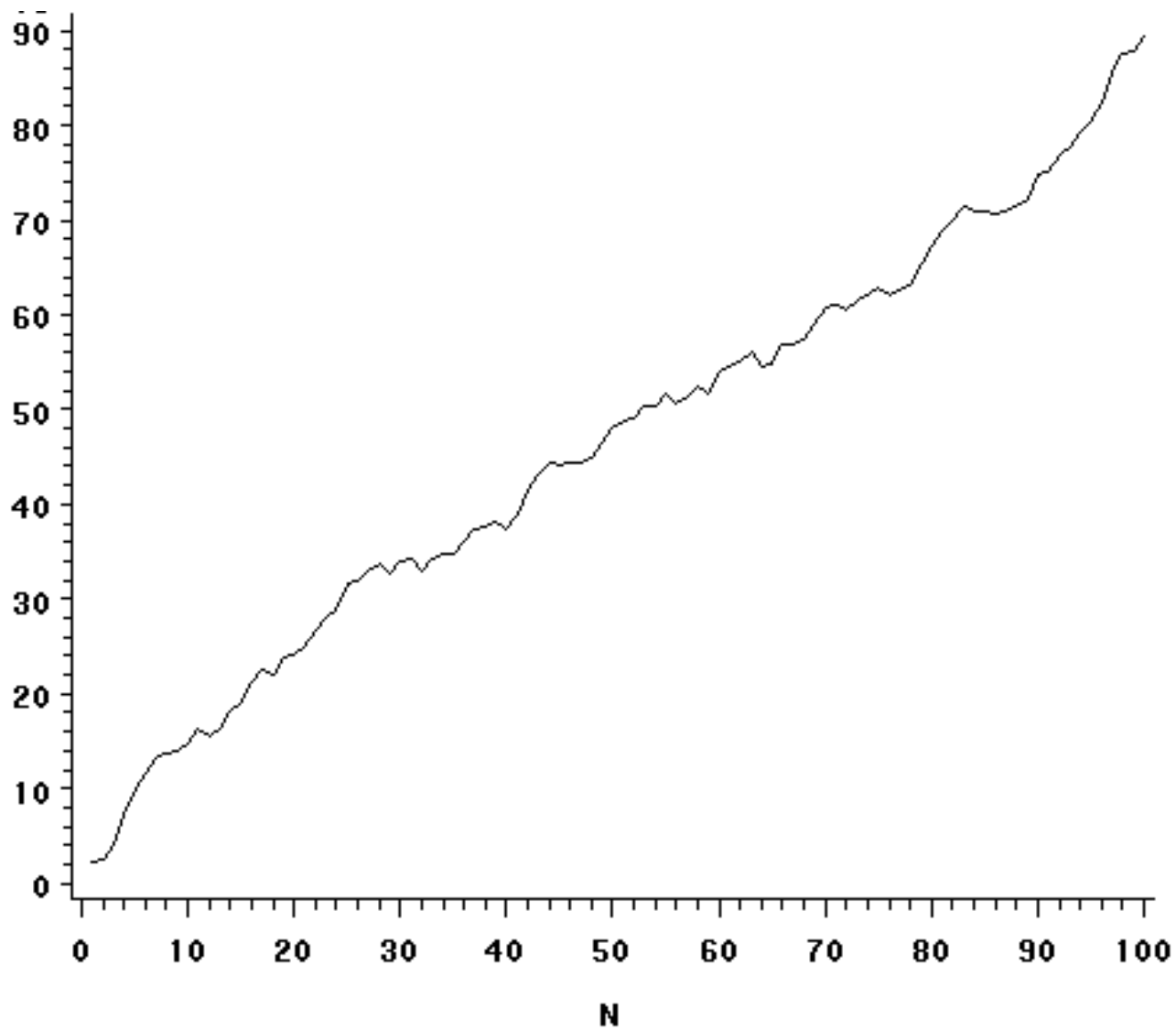
R - residual or random movement
(irregular variation)

The components of a time series are easily Identified and explained pictorially in a time series plot.

A time series plot is a sequence plot with the time series variable Y_t the vertical axis and time t on the horizontal axis.

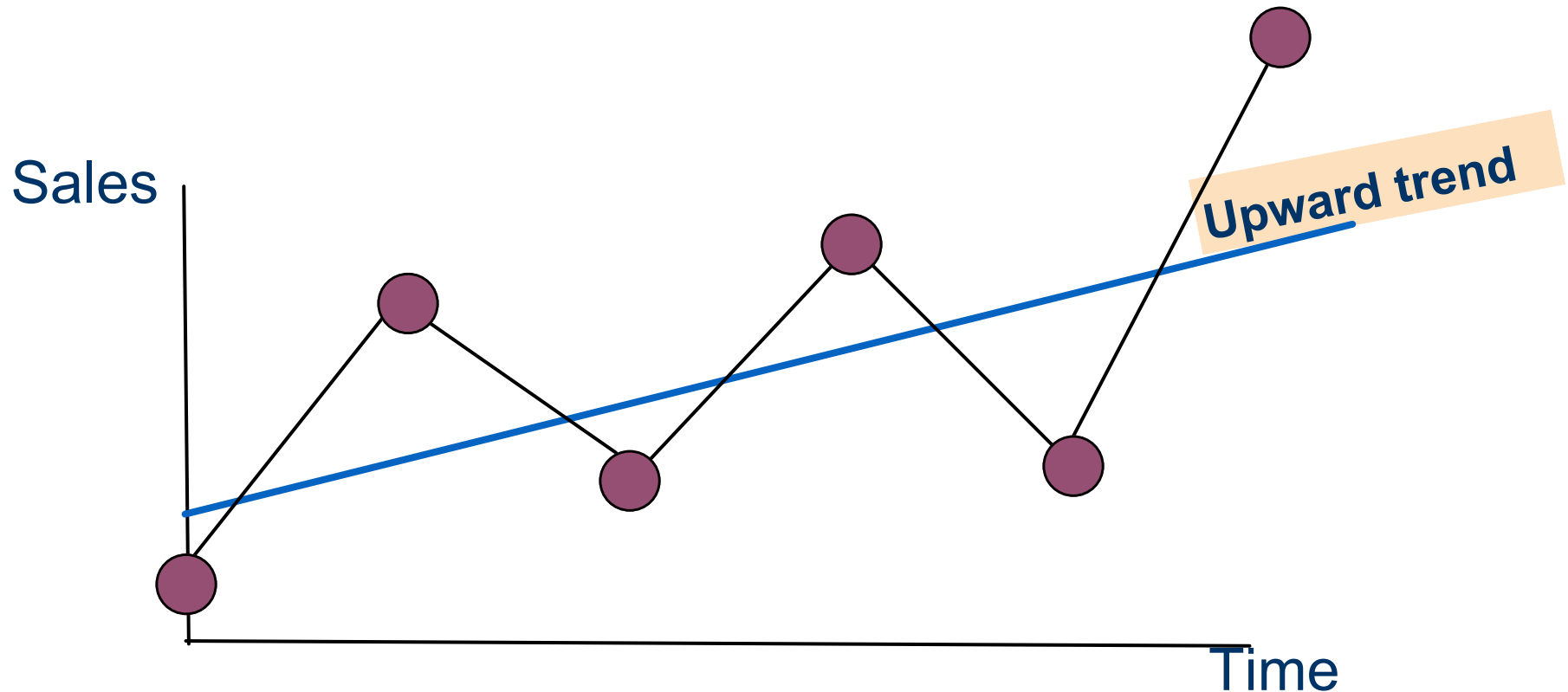
The figure 1.1 shows a trend in the time series values. The trend component describes the tendency of the value of the variable to increase or decrease over a long period of time.

Figure 1.1: Trend in a time series



Trend Component

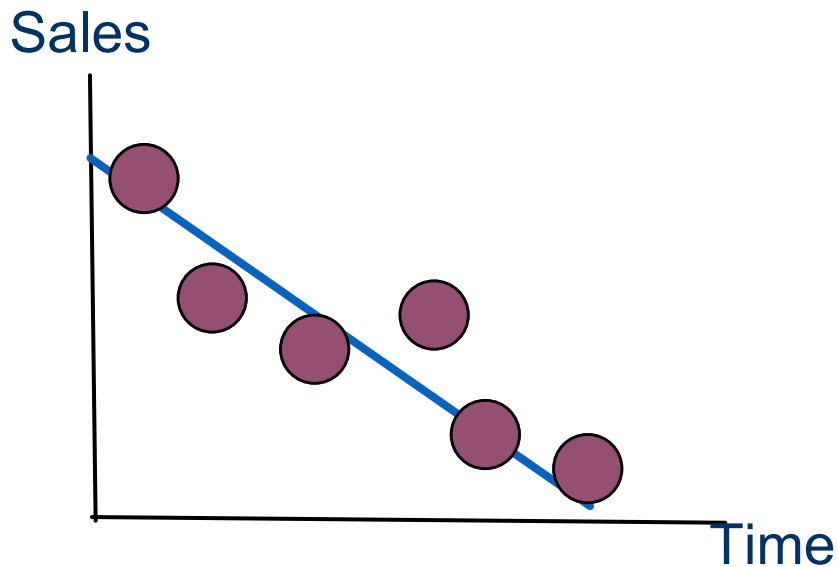
- Long-run increase or decrease over time (overall upward or downward movement)
- Data taken over a long period of time



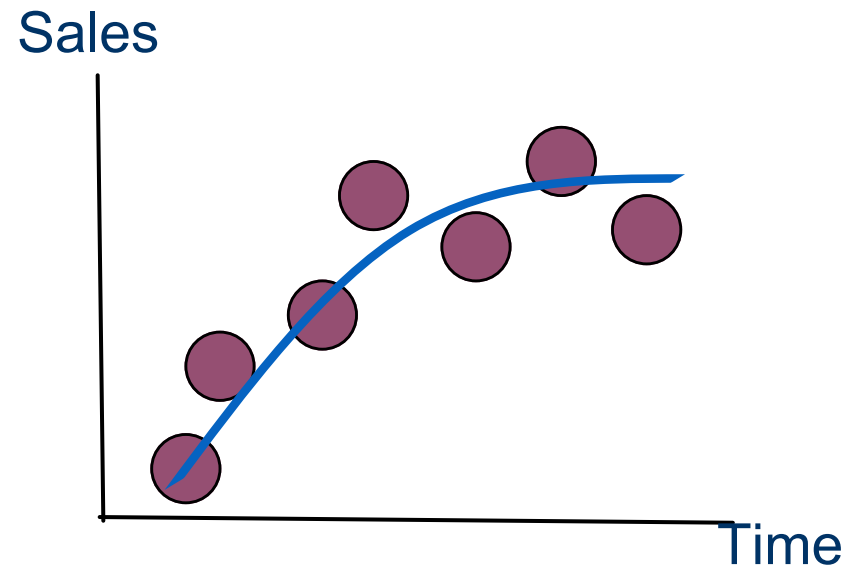
Trend Component

(continued)

- Trend can be upward or downward
- Trend can be linear or non-linear



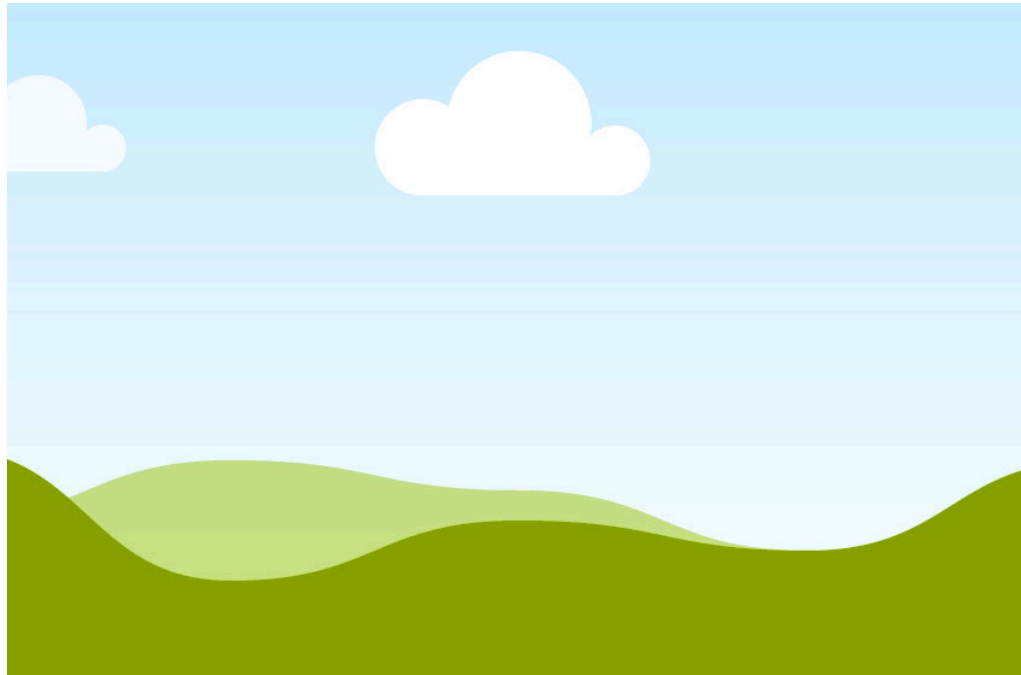
Downward linear trend



Upward nonlinear trend

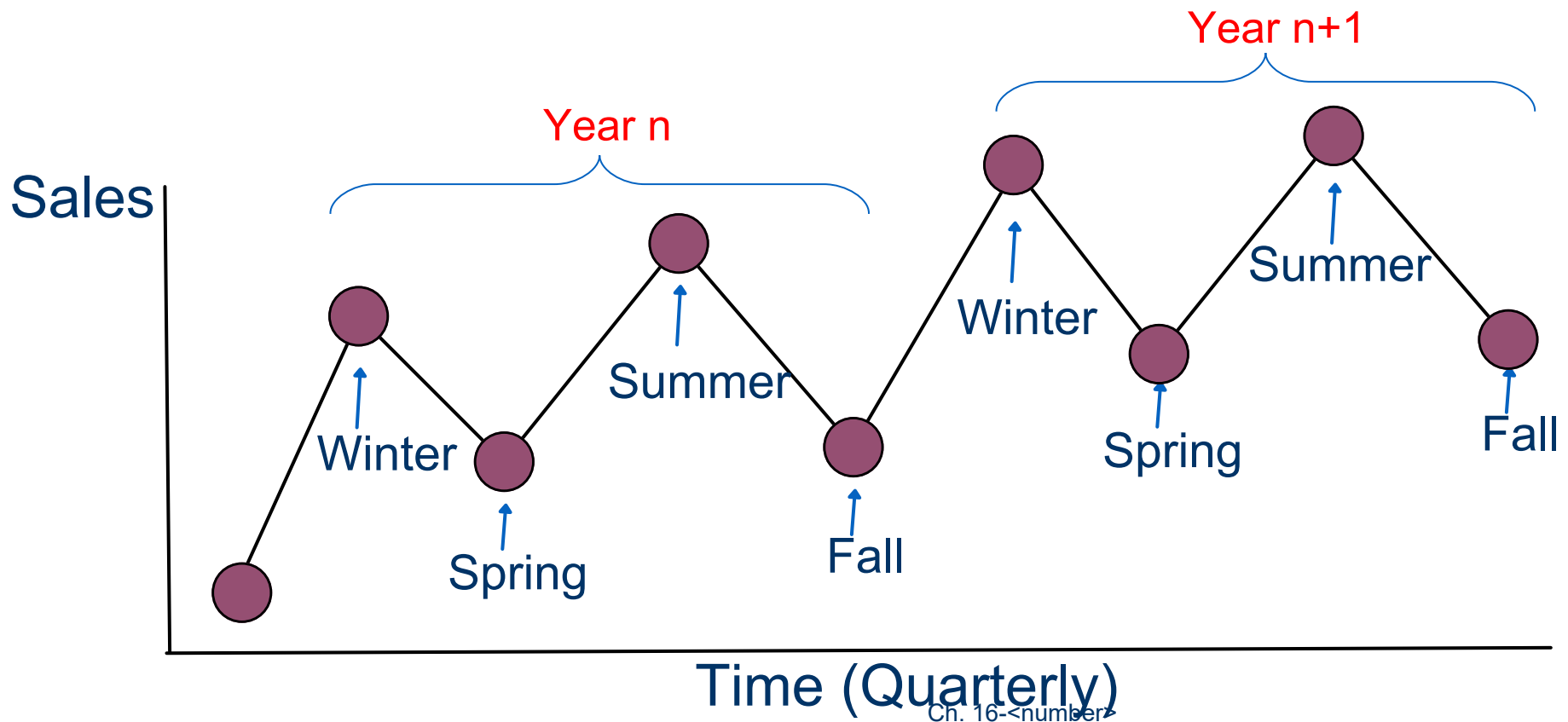
A seasonal variation in a series describes the fluctuations that recur during specific portions of each year (monthly or seasonally), the pattern of change in the time series within a year tends to be repeated from year to year producing a wavelike or oscillating curve.

Figure 1.4 : seasonal variations in a time series



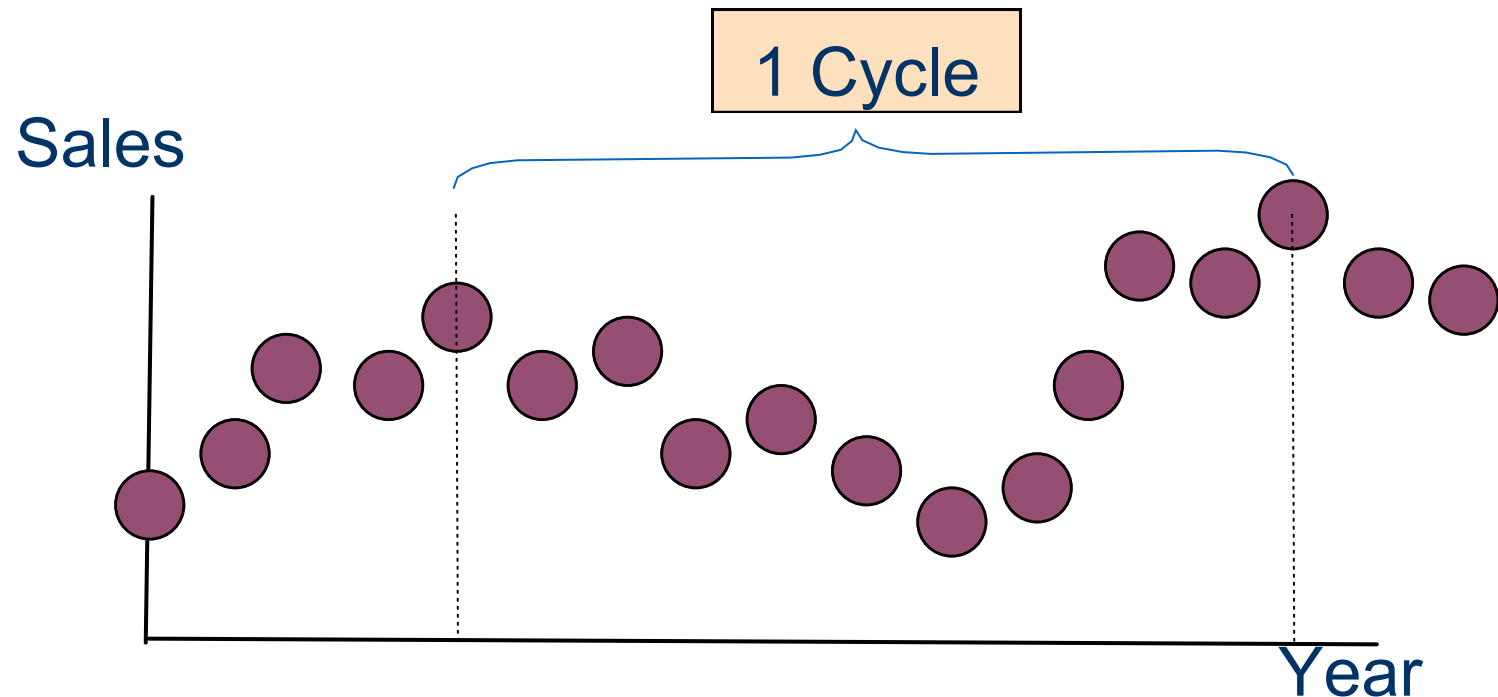
Seasonal Component

- Short-term regular wave-like patterns
- Observed within 1 year
- Often monthly or quarterly



Cyclical Component

- Long-term wave-like patterns
- Regularly occur but may vary in length
- Often measured peak to peak or trough to trough



A cyclical effect in a time series as shown in the figure 1.2 describes the fluctuation about the trend line.

In business the fluctuations are called business cycles.

figure 1.2 : cyclic variations in a time series



Irregular Component

- Unpredictable, random, “residual” fluctuations
- Due to random variations of
 - Nature
 - Accidents or unusual events
- “Noise” in the time series

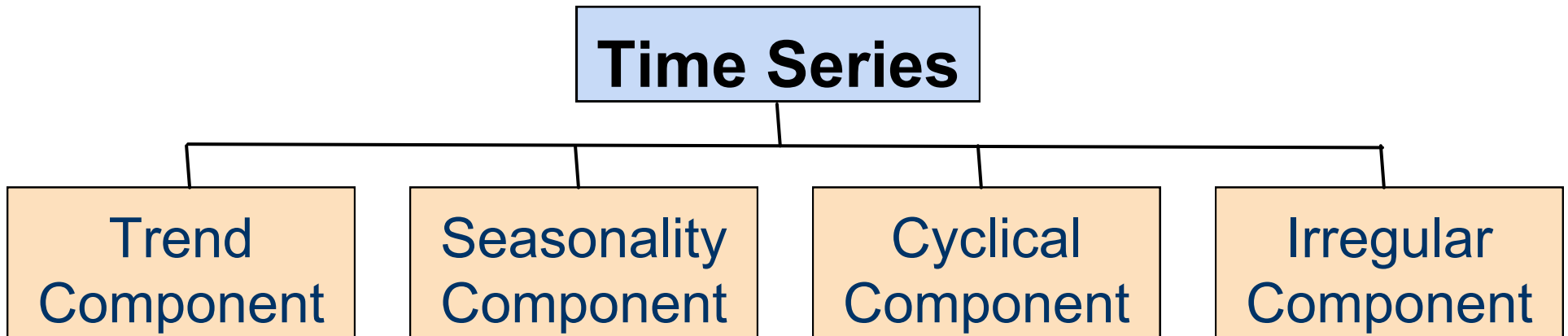
The final component, the residual effect is what remains after the trend, cyclical and seasonal components have been removed.

This component is not systematic and may be attributed to unpredictable influences.

Thus, the residual effect represents the random error component of a time series.

- One of the objectives of time series is to forecast some future values of the series.
- To obtain forecast some type of model that can be projected into the future must be used to describe the time series.
- One of the most widely used models is the additive model.

Time-Series Components



Trend Projection

- If a time series exhibits a linear trend, the method of least squares may be used to determine a trend line (projection) for future forecasts.
- Least squares, also used in regression analysis, determines the unique trend line forecast which minimizes the mean square error between the trend line forecasts and the actual observed values for the time series.
- The independent variable is the time period and the dependent variable is the actual observed value in the time series.

Linear Trend Regression

- Using the method of least squares, the formula for the trend projection is:

$$T_t = b_0 + b_1 t$$

where: T_t = linear trend forecast in period t

b_0 = intercept of the linear trend line

b_1 = slope of the linear trend line

t = time period

Linear Trend Regression

- For the trend projection equation $Tt = b_0 + b_1t$

$$b_1 = \frac{\sum_{t=1}^n (t - \bar{t})(Y_t - \bar{Y})}{\sum_{t=1}^n (t - \bar{t})^2}$$

$$b_0 = \bar{Y} - b_1\bar{t}$$

where: Y_t = value of the time series in period t
 n = number of time periods (observations)

$\bar{Y}$ = average values of the time series

$\bar{t}$ = average value of t

Linear Trend Regression

- Example: Auger's Plumbing Service

The number of plumbing repair jobs performed by Auger's Plumbing Service in the last nine months is listed on the right. Forecast the number of repair jobs Auger's will perform in December using the least squares method.

<u>Month</u>	<u>Jobs</u>	<u>Month</u>	<u>Jobs</u>
March	353	August	409
April	387	September	399
May	342	October	412
June	374	November	408
July	396		

Linear Trend Regression

(month) t	t	$t - \bar{t}$	$(t - \bar{t})^2$	Y_t	$(Y_t - \bar{Y})$	$(t - \bar{t})(Y_t - \bar{Y})$
(Mar.)	1	-4	16	353	-33.67	134.68
(Apr.)	2	-3	9	387	0.33	-0.99
(May)	3	-2	4	342	-44.67	89.34
(June)	4	-1	1	374	-12.67	12.67
(July)	5	0	0	396	9.33	0
(Aug.)	6	1	1	409	22.33	22.33
(Sep.)	7	2	4	399	12.33	24.66
(Oct.)	8	3	9	412	25.33	75.99
(Nov.)	9	4	16	408	21.33	85.32
Sum	45		60	3480		444.00

Linear Trend Regression

$$\bar{t} = 45/9 = 5 \quad \text{and} \quad \bar{Y} = 3480/9 = 386.667$$

$$b_1 = \frac{\sum_{t=1}^n (t - \bar{t})(Y_t - \bar{Y})}{\sum_{t=1}^n (t - \bar{t})^2} = \frac{3480}{60} = 7.12$$

$$b_0 = \bar{Y} - b_1 \bar{t} = 386.667 - 7.12(5) = 351.07$$

$$T_{10} = 351.07 + (7.12)(10) = \textcircled{422.27}$$

Trend Projection

- Example: Auger's Plumbing Service

Forecast for December (Month 10) using a three-period ($k = 3$) weighted moving average with weights of .6, .3, and .1 for the newest to oldest data, respectively. Then, compare this Month 10 weighted moving average forecast with the Month 10 trend projection forecast.

<u>Month</u>	<u>Jobs</u>	<u>Month</u>	<u>Jobs</u>
March	353	August	409
April	387	Septem.	399
May	342	October	412
June	374	Novem.	408
July	396		

Trend Projection

- Three-Month Weighted Moving Average

The forecast for December will be the weighted average of the preceding three months: September, October, and November.

$$\begin{aligned} F_{10} &= 0.1Y_{\text{Sep.}} + 0.3Y_{\text{Oct.}} + 0.6Y_{\text{Nov.}} \\ &= 0.1(399) + 0.3(412) + 0.6(408) \\ &= 408.3 \end{aligned}$$

- Trend Projection

$$F_{10} = 422.27 \text{ (from earlier slide)}$$

Nonlinear Trend Regression

- Sometimes time series have a curvilinear or nonlinear trend.
- A variety of nonlinear functions can be used to develop an estimate of the trend in a time series.
- One example is this quadratic trend equation:

$$T_t = b_0 + b_1t + b_2t^2$$

- Another example is this exponential trend equation:

$$T_t = b_0(b_1)^t$$

Nonlinear Trend Regression

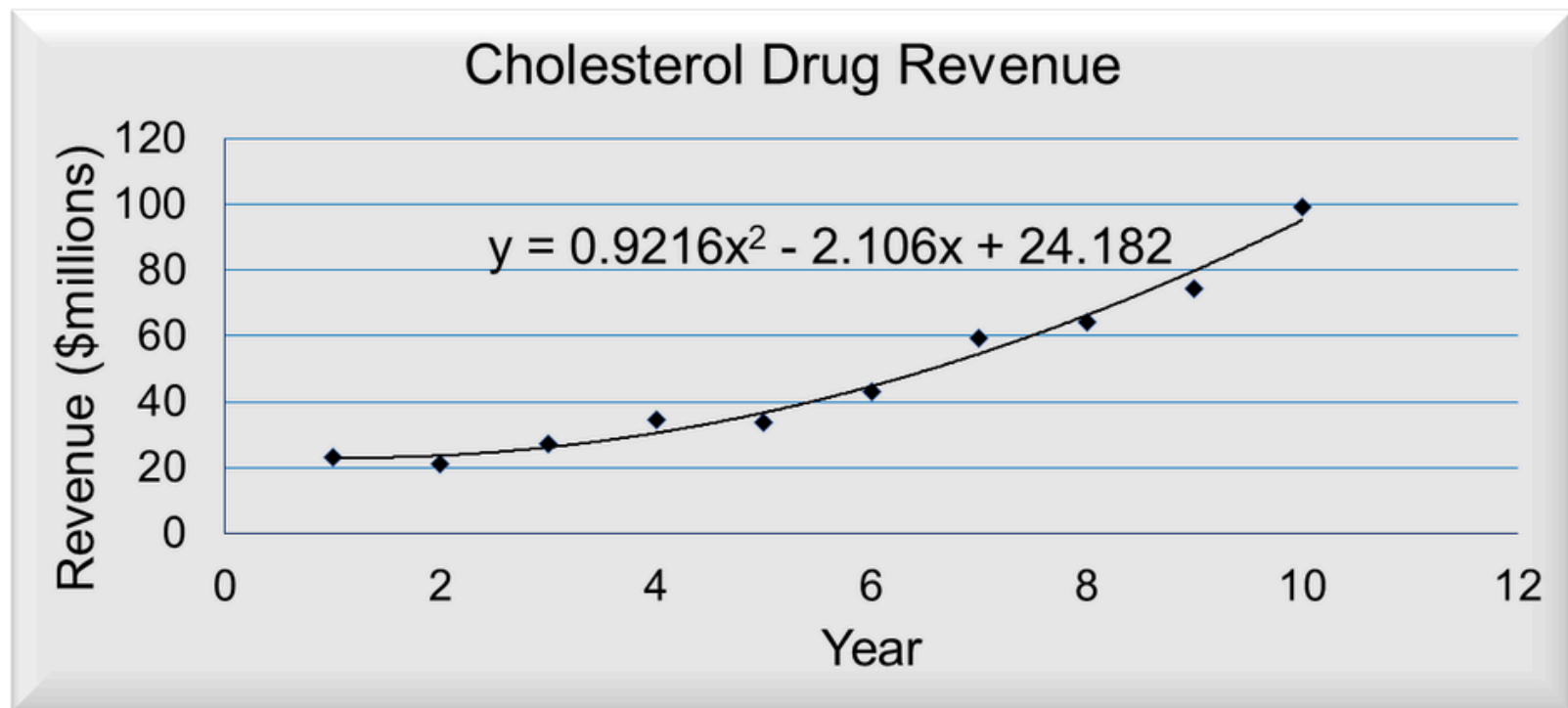
- Example: Cholesterol Drug Revenue

The annual revenue in millions of dollars for a cholesterol drug for the first 10 years of sales is shown below. A curvilinear function appears to be needed to model the long-term trend.

<u>Year</u>	<u>Revenue</u>	<u>Year</u>	<u>Revenue</u>
1	23.1	6	43.2
2	21.3	7	59.5
3	27.4	8	64.4
4	34.6	9	74.2
5	33.8	10	99.3

Nonlinear Trend Regression

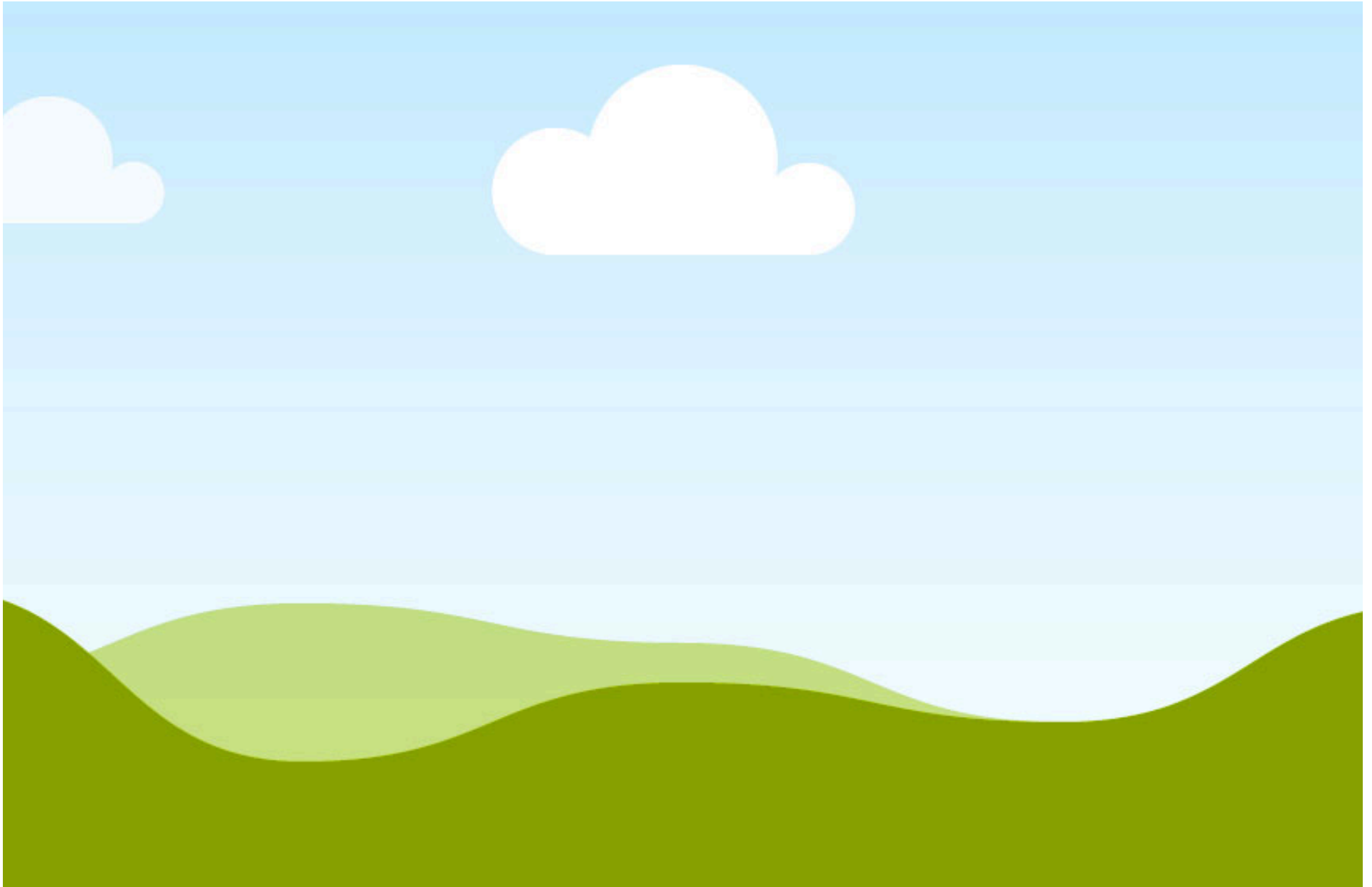
- Example: Cholesterol Drug Revenue



Filtering(Smoothing techniques)



Number of Militarized Interstate Disputes (MIDs), 1816–2001



Filtering(Smoothing techniques)

- A procedure to remove random variation revealing more clearly the underlying trend and cyclic components in a set of time series data, converting one time series to another by performing a linear operation for the use of forecasting.
- **Moving Average**
 - Smooth out (reduce) fluctuations and gives trend values to a fair degree of accuracy.
 - The average value of a number of adjacent time series values is taken as the trend value for the middle point.

Moving Averages: Smoothing the Time Series

- Calculate moving averages to get an overall impression of the pattern of movement over time
- This smooths out the irregular component

Moving Average: averages of a designated
number of consecutive
time series values

$(2m+1)$ -Point Moving Average

- A series of arithmetic means over time
- Result depends upon choice of m (the number of data values in each average)
- Examples:
 - For a 5 year moving average, $m = 2$
 - For a 7 year moving average, $m = 3$
 - Etc.
- Replace each x_t with

Moving Averages

- **Example:** Five-year moving average
 - First average:
 - Second average:
 - etc.

Example: Annual Data

Year	Sales
1	23
2	40
3	25
4	27
5	32
6	48
7	33
8	37
9	37
10	50
11	40
etc...	etc...

...

...

Calculating Moving Averages

- Let $m = 2$

Year	Sales
1	23
2	40
3	25
4	27
5	32
6	48
7	33
8	37
9	37
10	50
11	40

etc...

Average Year	5-Year Moving Average
3	29.4
4	34.4
5	33.0
6	35.4
7	37.4
8	41.0
9	39.4
...	...

- Each moving average is for a consecutive block of $(2m+1)$ years

Annual vs. Moving Average

- The 5-year moving average smoothes the data and shows the underlying trend

Centered Moving Averages

(continued)

- Let the time series have period s , where s is even number
 - i.e., $s = 4$ for quarterly data and $s = 12$ for monthly data
- To obtain a centered s -point moving average series X_t^* :
 - Form the s -point moving averages
 - Form the centered s -point moving averages

Centered Moving Averages

- Used when an even number of values is used in the moving average
- Average periods of 2.5 or 3.5 don't match the original periods, so we average two consecutive moving averages to get **centered moving averages**

Average Period	4-Quarter Moving Average	Centered Period	Centered Moving Average
2.5	28.75	3	29.88
3.5	31.00	4	32.00
4.5	33.00	5	34.00
5.5	35.00	6	36.25
6.5	37.50	7	38.13
7.5	38.75	8	39.00
8.5	39.25	9	40.13
9.5	41.00		

etc...

Moving Averages and Exponential Smoothing

- Moving Averages
 - Weighted Moving Averages
 - Exponential Smoothing
-
- They are called smoothing methods because their objective is to smooth out the random fluctuations in the time series.
 - They are most appropriate for short-range forecasts.

Moving Averages Forecasting

- The moving averages method uses the average of the most recent k data values in the time series as the forecast for the next period.

$$F_{t+1} = \frac{\sum(\text{most recent } k \text{ data values})}{k} = \frac{Y_t + Y_{t-1} + \cdots + Y_{t-k+1}}{k}$$

where: F_{t+1} = forecast of the time series for period $t + 1$

- Each observation in the moving average calculation receives the s

Moving Averages

- To use moving averages to forecast, we must first select the order k , or number of time series values, to be included in the moving average.
- A smaller value of k will track shifts in a time series more quickly than a larger value of k .
- If more past observations are considered relevant, then a larger value of k is better.

Moving Averages

- Example: Rosco Drugs

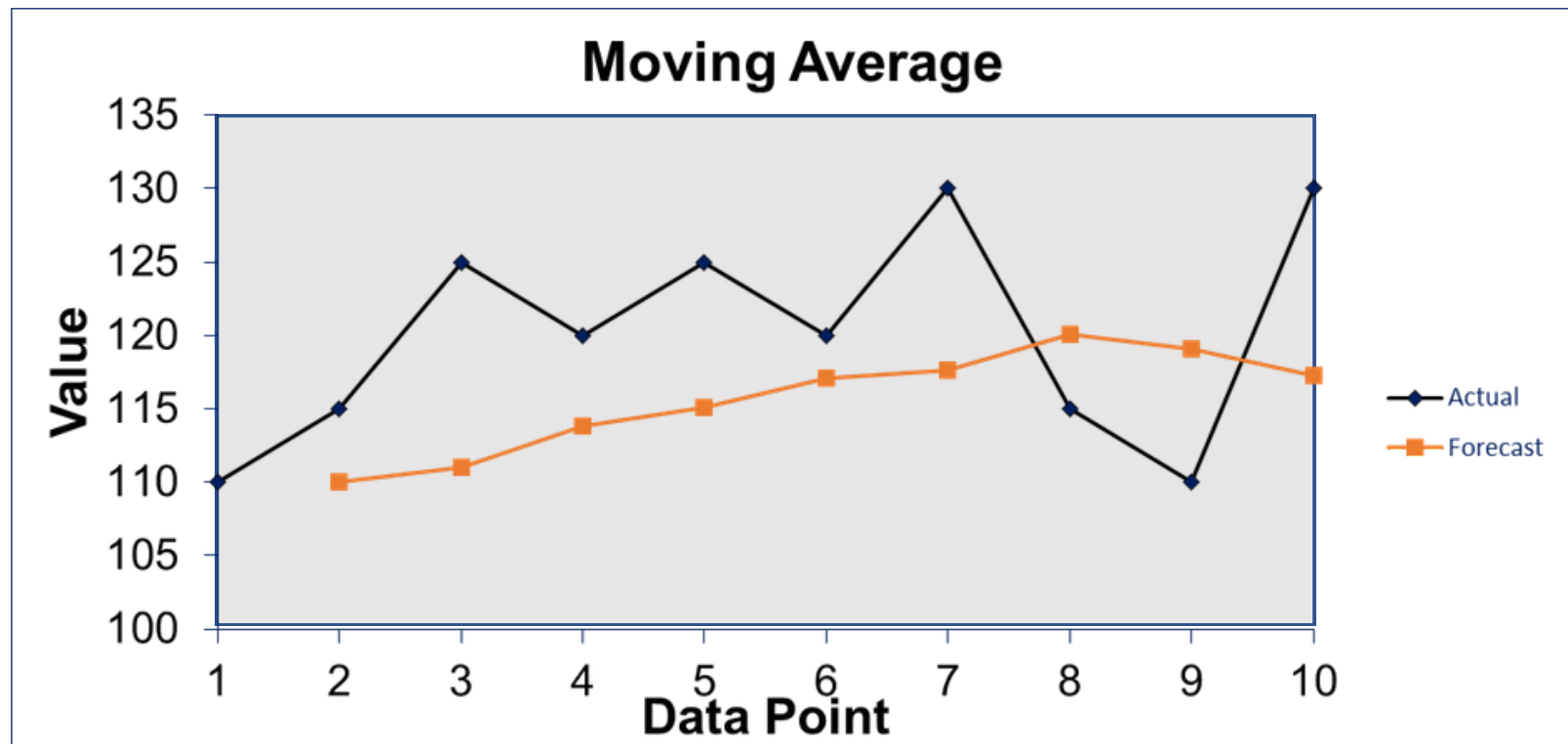
If Rosco Drugs uses a 3-period moving average to forecast sales, what are the forecasts for weeks 4-11?

Week	Sales		Week	Sales
1	110		6	120
2	115		7	130
3	125		8	115
4	120		9	110
5	125		10	130

Moving Averages

<u>Week</u>	<u>Sales</u>	<u>3MA Forecast</u>
1	110	
2	115	
3	125	
4	120	116.7
5	125	120.0
6	120	123.3
7	130	121.7
8	115	125.0
9	110	121.7
10	130	118.3
11		118.3

Moving Averages



Moving Averages

<u>Week</u>	<u>Sales</u>	3MA <u>Forecast</u>	<u>Forecast</u> <u>Error</u>	<u>Absolute</u> <u>Error</u>	<u>Squared</u> <u>Error</u>	<u>Abs.%</u> <u>Error</u>
1	110					
2	115					
3	125					
4	120	116.7	3.3	3.3	10.89	2.75
5	125	120.0	5.0	5.0	25.00	4.00
6	120	123.3	-3.3	3.3	10.89	2.75
7	130	121.7	8.3	8.3	68.89	6.38
8	115	125.0	-10.0	10.0	100.00	8.70
9	110	121.7	-11.7	11.7	136.89	10.64
10	130	118.3	11.7	11.7	136.89	9.00
		<u>Total</u>	6.6	53.3	489.45	44.22

Moving Averages

- 3-MA Forecast Accuracy

$$\text{MAE} = \frac{53.3}{7} = 7.61$$

$$\text{MSE} = \frac{489.45}{7} = 69.92$$

$$\text{MAPE} = \frac{44.22}{7} = 6.32\%$$

The 3-week moving average approach provided more accurate forecasts than the naïve approach.

Weighted Moving Averages

- Weighted Moving Averages
 - To use this method, we must first select the number of data values to be included in the average.
 - Next, we must choose the weight for each of the data values.
 - The more recent observations are typically given more weight than older observations.
 - For convenience, the weights should sum to 1.

Weighted Moving Averages

- Weighted Moving Averages
 - An example of a 3-period weighted moving average (3WMA) is
- $$3WMA = 0.2(110) + 0.3(115) + 0.5(125) = 119$$
- In this example the weights are 0.2, 0.3, and 0.5 (which sum to 1.0)
 - 125 is the most recent of the three observations.

Exponential Smoothing

- This method is a special case of a weighted moving averages method; we select only the weight for the most recent observation.
- The weights for the other data values are computed automatically and become smaller as the observations grow older.
- The exponential smoothing forecast is a weighted average of all the observations in the time series.
- The term *exponential smoothing* comes from the exponential nature of the weighting scheme for the historical values.

Exponential Smoothing

- Exponential Smoothing Forecast

$$F_{t+1} = Y_t + (1 - \alpha)F_t$$

where:

F_{t+1} = forecast of the time series for period $t + 1$

Y_t = actual value of the time series in period t

F_t = forecast of the time series for period t

α = smoothing constant ($0 \leq \alpha \leq 1$)

and let:

$F_2 = Y_1$ (to initiate the computations)

Exponential Smoothing

- Exponential Smoothing Forecast
 - With algebraic manipulation, we can rewrite

$$F_{t+1} = Y_t + (1 - \alpha)F_t \text{ as:}$$

$$F_{t+1} = F_t + \alpha(Y_t - F_t)$$

- We see that the new forecast F_{t+1} is equal to the previous forecast F_t plus an adjustment, which is α times the most recent forecast error, $Y_t - F_t$.

Exponential Smoothing

- Example: Rosco Drugs

If Rosco Drugs uses exponential smoothing to forecast sales, which value for the smoothing constant α , 0.1 or 0.8, gives better forecasts?

Week	Sales		Week	Sales
1	110		6	120
2	115		7	130
3	125		8	115
4	120		9	110
5	125		10	130

Exponential Smoothing

- Using Smoothing Constant Value = .1

$$F_2 = Y_1 = 110$$

$$F_3 = .1Y_2 + .9F_2 = .1(115) + .9(110) = 110.50$$

$$F_4 = .1Y_3 + .9F_3 = .1(125) + .9(110.5) = 111.95$$

$$F_5 = .1Y_4 + .9F_4 = .1(120) + .9(111.95) = 112.76$$

$$F_6 = .1Y_5 + .9F_5 = .1(125) + .9(112.76) = 113.98$$

$$F_7 = .1Y_6 + .9F_6 = .1(120) + .9(113.98) = 114.58$$

$$F_8 = .1Y_7 + .9F_7 = .1(130) + .9(114.58) = 116.12$$

$$F_9 = .1Y_8 + .9F_8 = .1(115) + .9(116.12) = 116.01$$

$$F_{10} = .1Y_9 + .9F_9 = .1(110) + .9(116.01) = 115.41$$

Exponential Smoothing

- Using Smoothing Constant Value = .8

$$F_2 = \quad \quad \quad = 110$$

$$F_3 = .8(115) + .2(110) = 114.00$$

$$F_4 = .8(125) + .2(114) = 122.80$$

$$F_5 = .8(120) + .2(122.80) = 120.56$$

$$F_6 = .8(125) + .2(120.56) = 124.11$$

$$F_7 = .8(120) + .2(124.11) = 120.82$$

$$F_8 = .8(130) + .2(120.82) = 128.16$$

$$F_9 = .8(115) + .2(128.16) = 117.63$$

$$F_{10} = .8(110) + .2(117.63) = 111.53$$

Exponential Smoothing ($\alpha = .1$)

<u>Week</u>	<u>Sales</u>	$\alpha = .1$ <u>Forecast</u>	<u>Forecast Error</u>	<u>Absolute Error</u>	<u>Squared Error</u>	<u>Abs.% Error</u>
1	110					
2	115	110.00	5.00	5.00	25.00	4.35
3	125	110.50	14.50	14.50	210.25	11.60
4	120	111.95	8.05	8.05	64.80	6.71
5	125	112.76	12.24	12.24	149.94	9.79
6	120	113.98	6.02	6.02	36.25	5.02
7	130	114.58	15.42	15.42	237.73	11.86
8	115	116.12	-1.12	1.12	1.26	0.97
9	110	116.01	-6.01	6.01	36.12	5.46
10	130	115.41	14.59	14.59	212.87	11.22
<u>Total</u>				82.95	974.22	66.98

Exponential Smoothing ($\alpha = .1$)

- Forecast Accuracy

$$\text{MAE} = \frac{82.95}{9} = 9.22$$

$$\text{MSE} = \frac{974.22}{9} = 108.25$$

$$\text{MAPE} = \frac{66.98}{9} = 7.44\%$$

Exponential smoothing (with $\alpha = .1$) provided less accurate forecasts than the 3-MA approach.

Exponential Smoothing ($\alpha = .8$)

<u>Week</u>	<u>Sales</u>	$\alpha = .8$ <u>Forecast</u>	<u>Forecast Error</u>	<u>Absolute Error</u>	<u>Squared Error</u>	<u>Abs.% Error</u>
1	110					
2	115	110.00	5.00	5.00	25.00	4.35
3	125	114.00	11.00	11.00	121.00	8.80
4	120	122.80	-2.20	2.20	7.84	1.83
5	125	120.56	4.44	4.44	19.71	3.55
6	120	124.11	-4.11	4.11	16.91	3.43
7	130	120.82	9.18	9.18	84.23	7.06
8	115	128.16	-13.16	13.16	173.30	11.44
9	110	117.63	-7.63	7.63	58.26	6.94
10	130	111.53	18.47	18.47	341.27	14.21
			<u>Total</u>	75.19	847.52	61.61

Exponential Smoothing ($\alpha = .8$)

- Forecast Accuracy

$$\text{MAE} = \frac{75.19}{9} = 8.35$$

$$\text{MSE} = \frac{847.52}{9} = 94.17$$

$$\text{MAPE} = \frac{61.61}{9} = 6.85\%$$

Exponential smoothing (with $\alpha = .8$) provided more accurate forecasts than ES with $\alpha = .1$, but less accurate than the moving average (with $k = 3$).

THANK YOU